Data retention and deletion

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

List the copies. Apply the boundary. Verify the stated scope.

Trace fictional retention and removal across declared stores while preserving only permitted metadata and incomplete-deletion evidence.

Keep one habit: Policy/clock contract → complete copy ledger → explicit backup picture → residual/audit probes → copied-store simulation → independent 4-day rule.

MAIN LESSON · SECTION 1

Apply the age boundary to every declared content copy

> Deleting the visible row is not proof that the listed cache or backup copy is gone.

Learning goals

I can explain the **mechanism** and terminology of data retention and deletion.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve privacy-aware records and declared scope

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Sandboxing and resource limits. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
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> _____
>
> _____

End goal

Explain data retention and deletion using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of data retention and deletion

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the inventory, age rule and separate audit policy

> A scoped deletion claim must say which copies were actually checked.

Begin with primary and cache copies

A transcript may have a primary copy and a cache copy. If the fictional policy says to remove content at age 7 days or older, deleting just the primary copy leaves the cache out of compliance. First list the stores covered by the policy, then apply the same boundary to each relevant copy and verify the remaining state. Preserve only allowed audit metadata, not the expired transcript under a new label.

Define retention, deletion and residual copies

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Complete fictional policy and age clocks

Main fictional retention policy	Exact classroom rule
Age measurement	Supplied nonnegative whole-day ages at one fixed snapshot
Session-content boundary	Age at least 7 days: removal due; age below 7: retain for now

Main fictional retention policy	Exact classroom rule
Main listed content stores	Primary and cache, both initially holding the three supplied records
Backups in main case	Need their separately stated expiry process; not inspected by this fixture
Audit metadata	May remain under a separate 30 day policy, without transcript content
Audit age clock	Age since the audit event was created, not the original transcript age
Verification	Check due content absent from every listed store in the modeled scope
Code scope	Transform copies of fictional dictionaries; no actual storage deletion or secure erasure
Policy status	These toy lifetimes are supplied exercise rules, not legal retention requirements

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace each record and each declared copy

> The content and audit clocks must not be silently substituted for each other.

Original 7-day content/30day metadata mechanism

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Complete age-by-copy action ledger

Main record ID	Content age	Primary/cache copies initially	Required modeled action
S6	6	One in each of2 stores	Retain both content copies for now
S7	7	One in each of2 stores	Remove both: equality reaches deletion boundary
S9	9	One in each of2 stores	Remove both: above boundary
After successful scoped simulation	S6 remains in both stores	Four due entries removed	Only primary/cache postconditions checked
Main backups	No snapshot supplied	Separate expiry policy still needed	Do not claim global deletion
New allowed audit events	Event age 0	No transcript payload	Audit age 29: retain; age 30: due

State scope and boundary before judging

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Data retention and deletion — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Record ages are 6, 7, 9 days. Cache contains copies of all three.	Toy policy retains session content for 7 days and deletes it when age is at least 7.	Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Change a removal outcome and preserve the failure record

> An outstanding required copy makes scoped deletion incomplete.

Current task and response

- Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the failed pictured backup removal

- Change: Primary deletion succeeds but backup deletion fails.

▼ Check after predicting

Prediction: Record incomplete deletion and follow the retry or escalation procedure.

Reason: One required copy remains in the declared scope.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Compare successful and incomplete deletion within the stated inventory

> Do not borrow an unspecified main backup policy for the picture’s explicit backup rule.

Main age inputs and primary/cache conclusion

Record ages are 6, 7, 9 days. Cache contains copies of all three.

Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Preserve metadata without expired content

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only policy-permitted audit metadata without expired content so a later revision can be compared honestly.

Separate pictured inventory and inclusive age boundary

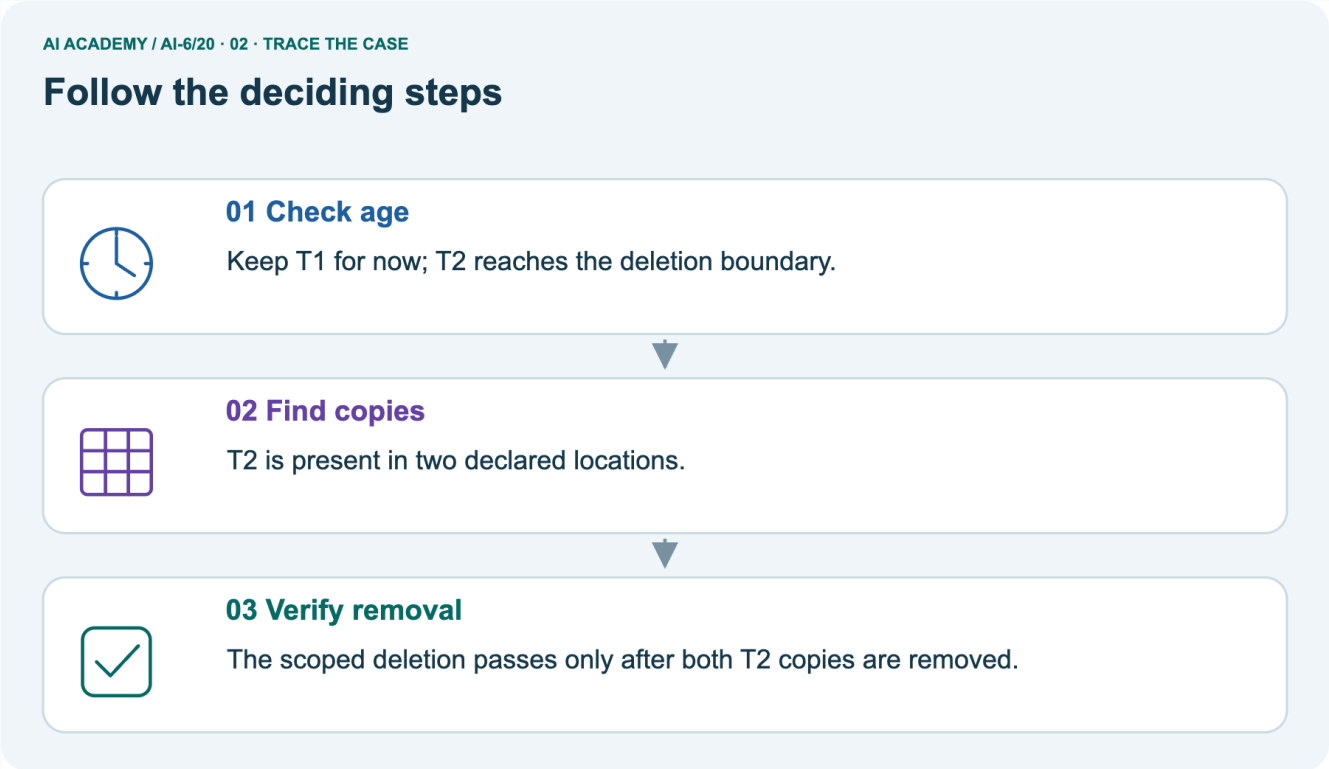
Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

T1 is six days old. T2 is seven days old with one backup. Scope lists primary storage and that backup only.

The main example checks primary/cache and leaves backup expiry unverified. The picture has a separate explicit scope: primary storage and the one listed backup of T2, both covered by its fictional 7-day content policy. T1 at age 6 is retained; T2 at age 7 requires removal from two locations. If primary removal succeeds but the backup remains, record incomplete deletion and that residual location.

A separate retry can clear the remaining modeled copy while the failed report stays in the history. This establishes no removal from unknown locations or real physical media.

Trace the two required T2 locations



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check age	Keep T1 for now; T2 reaches the deletion boundary.
2. Find copies	T2 is present in two declared locations.
3. Verify removal	The scoped deletion passes only after both T2 copies are removed.

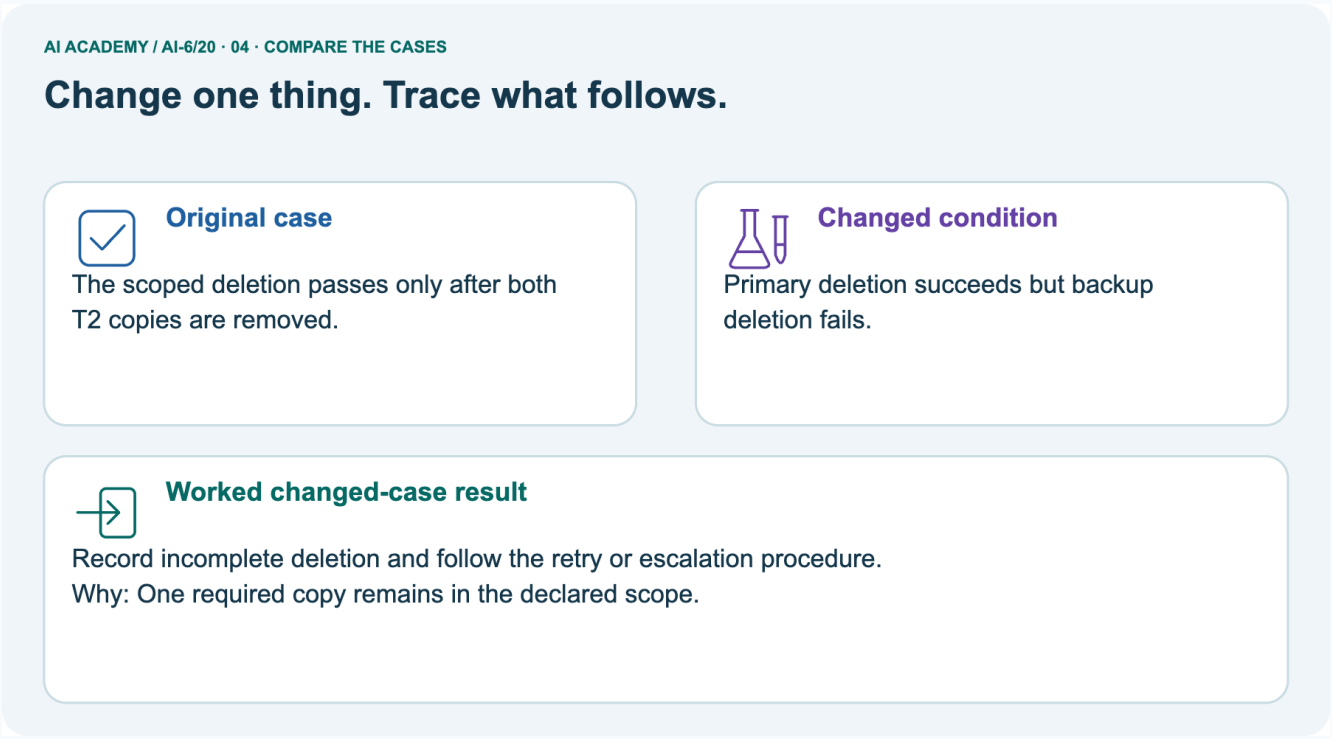
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

Predict primary success with backup failure

- Change one thing: Primary deletion succeeds but backup deletion fails. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain verification scope and unknown-copy limits

- The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

- Read the labelled picture: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

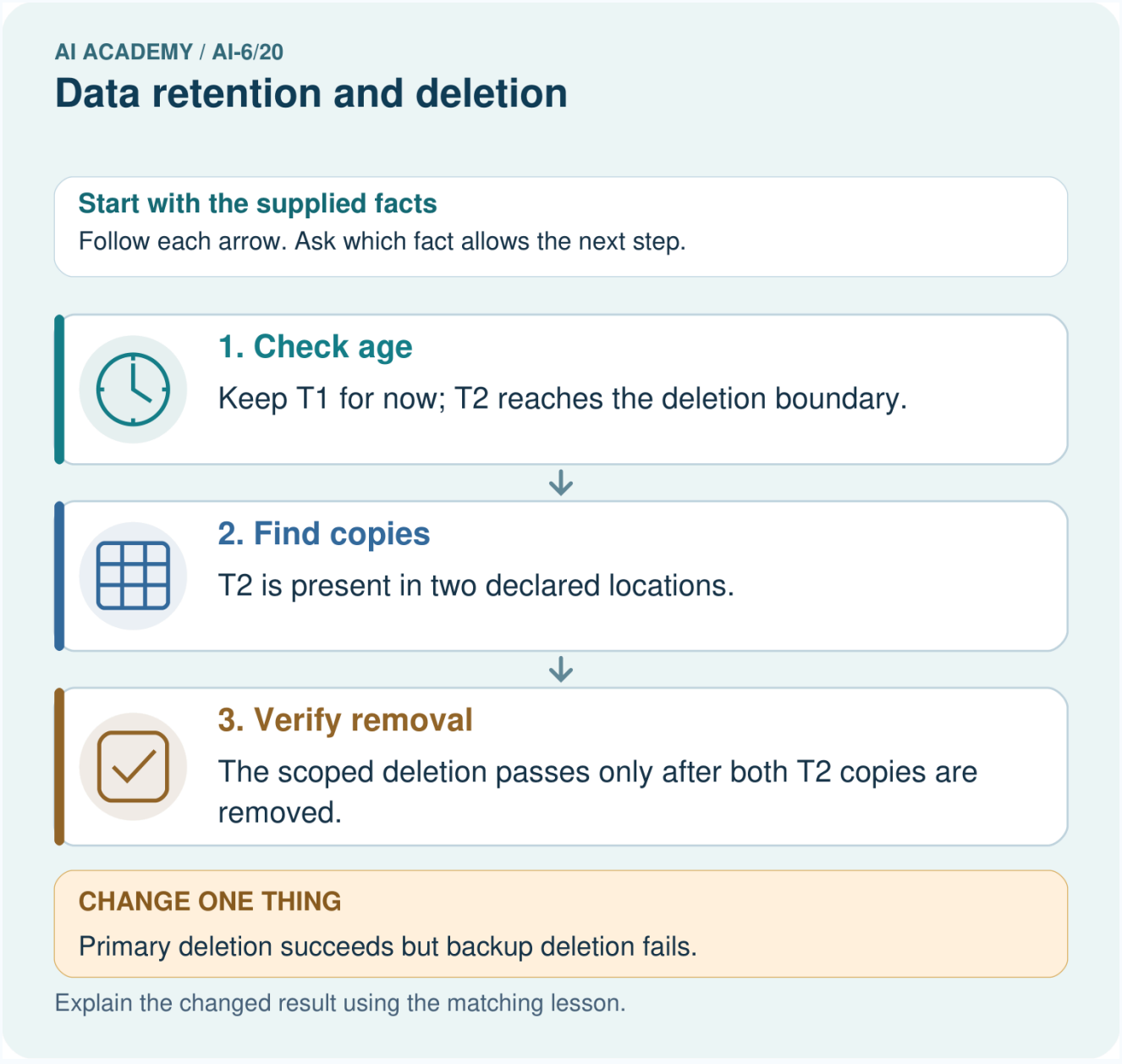
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Data retention and deletion

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep retention scope, metadata lifetime and verification claims precise

> A toy age rule does not establish a real legal obligation or erasure guarantee.

Core fictional-policy qualification and error analysis

This fictional retention exercise is not a statement of legal retention requirements.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete age, inventory and audit-content probes

Retention or inventory probe	Expected interpretation
Content age 6 under7 day rule	Retain for now
Content age 7or9	Removal due from every applicable listed content store
Primary removed, listed backup remains	Scoped deletion incomplete; do not report success
Unknown record age in a listed store	Reject incomplete manifest rather than guess retention status
No listed-store inventory	Reject empty/incomplete fixture; do not claim vacuous global deletion
New content-free audit event	Own age clock starts at creation; its 30 day lifetime is separate
Expired content copied into audit payload	Violates the stated no-content metadata policy

Retention or inventory probe	Expected interpretation
Same modeled removal retried	Check remaining copies and retain prior failures; do not overwrite history
All listed copies absent in the model	Scoped postcondition only, not proof of unknown-store deletion or physical erasure

Guided explanation and repair

- A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise the boundary and the remaining-copy check

> Name both the due record and every location the policy covers.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Data retention and deletion”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual policy-scope check

- Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Backup-picture response

- Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

> **My explanation using a pictured step and the given facts**

Reserve the selected 4-day policy and residual-log case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete scoped-retention and incomplete-removal simulation

> Inspect the transformed mock inventory, residual copies and content-free audit record.

Current 6/7/9-day content-copy investigation

Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

Complete runnable copy transformation and failure/retry checks

```
"""Transform copies of fictional store dictionaries; no real deletion/erasure."""
from copy import deepcopy

def simulate_retention(ages,stores,content_days=7,failed_removals=frozenset()):
    if type(ages) is not dict or not ages or type(stores) is not dict or not stores:
        raise ValueError('nonempty age manifest and listed stores required')
    if any(type(location) is not str or not location or type(items) is not dict for
location,items in stores.items()):
        raise ValueError('invalid listed store schema')
    if any(type(record_id) is not str or not record_id for record_id in ages):
        raise ValueError('record IDs required')
    if type(content_days) is not int or content_days<=0:
        raise ValueError('positive content lifetime required')
    if any(type(age) is not int or age<0 for age in ages.values()):
        raise ValueError('nonnegative whole-day fixture ages required')
    if any(record_id not in ages for items in stores.values() for record_id in items):
        raise ValueError('listed copy missing age metadata')
    after=deepcopy(stores)
    due=[record_id for record_id,age in ages.items() if age>=content_days]
    removed=[];failed=[]
    for record_id in due:
        for location,items in after.items():
            if record_id not in items:continue
            pair=(location,record_id)
            if pair in failed_removals:failed.append(pair)
            else:
                items.pop(record_id)  # simulated dictionary update only
                removed.append(pair)
    residuals=[(location,record_id) for location,items in after.items()
               for record_id in due if record_id in items]
    report={'scope':sorted(after),'due_ids':due,'removed':removed,'failed':failed,
```

```
            'residuals':residuals,'scoped_complete':not residuals}
    # Separate audit-event clock: these new content-free events have age zero.
    audit=[{'record_id':record_id,'action':'removed_in_mock',
            'location':location,'event_age_days':0} for location,record_id in removed]
    return after,report,audit

def retain_audit(age_days):
    if type(age_days) is not int or age_days<0:raise ValueError('invalid audit-event age')
    return age_days<30

ages={'S6':6,'S7':7,'S9':9}
primary={'S6':'fictional transcript6','S7':'fictional transcript7','S9':'fictional
transcript9'}
stores={'primary':dict(primary),'cache':dict(primary)}
before=deepcopy(stores)
after,report,audit=simulate_retention(ages,stores)
assert all(set(items)=={'S6'} for items in after.values())
assert len(report['removed'])==4 and report['scoped_complete'] and stores==before
assert all(set(r)=={'record_id','action','location','event_age_days'} for r in audit)
assert retain_audit(29) and not retain_audit(30)
print('Main modeled primary/cache outcome:',report)
print('Main remaining IDs:',{name:sorted(items) for name,items in after.items()})
print('Unverified scope: backups require their separately stated expiry process; no
global-deletion claim.')
print('New content-free audit events:',audit,'audit boundary
29/30:',retain_audit(29),retain_audit(30))
# Separate picture explicitly requires removal from primary AND its listed backup.
picture_ages={'T1':6,'T2':7}
picture_stores={'primary':{'T1':'fictional T1','T2':'fictional T2'},
                'backup':{'T2':'fictional T2'}}
picture_after,picture_report,_=simulate_retention(picture_ages,picture_stores)
failed_after,failed_report,_=simulate_retention(picture_ages,picture_stores,
                                                failed_removals={('backup','T2')})
assert picture_report['scoped_complete'] and len(picture_report['removed'])==2
assert not failed_report['scoped_complete'] and failed_report['residuals']==
[('backup','T2')]
retry_after,retry_report,_=simulate_retention(picture_ages,failed_after)
assert retry_report['scoped_complete'] and failed_report['residuals']==[('backup','T2')]
print('Picture all listed removals:',picture_report)
print('Picture backup removal fails:',failed_report)
print('Separate retry simulation:',retry_report)
print('Only copied mock dictionaries changed; original fixtures remain. No file, database,
log, backup or memory is securely erased.')
```

Expected scoped outcomes, separate clocks and limits

1. Predict due IDs, removed-copy count and residuals before running or tracing. The main simulation removes four S7/S9 entries from copied primary/cache dictionaries and retains S6; backups remain outside that verified scope. It emits content-free audit events with their own age 0 clock and checks the 30 day metadata boundary.

   The separate picture removes T2 from primary/backup, then models backup failure and a later retry without replacing the failed report. Original fictional input dictionaries remain for the classroom replay; this is a model transformation, not actual deletion of records or secure memory/media erasure. No files, databases, live logs or backups are changed.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply a new inclusive 4-day boundary across a temporary log

> Use the newly supplied policy rather than the main 7-day example.

Complete selected independent X/Y/Z and residual-copy task

Fictional retention policy deletes records at age >=4 days. X is3 days, Y4, Z5. A copy of Y remains in a temporary log after its main record is deleted. Evaluate compliance.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Attempt this without hints

Retention is5 days, inclusive **deletion boundary**. Ages 4, 5, 6; a log stores complete content. Trace required action.

> **My independent procedure, result and limitation**
>
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save policy versions, store inventory and residual-copy evidence

> Keep permitted audit metadata while excluding expired content from the diagnostic artifact.

Project task and artifact

Add a labelled data retention and deletion **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of data retention and deletion.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a scoped verification claim and an explicit residual if needed

> Allowed metadata is not a way to keep expired content indefinitely.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of data retention and deletion

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Retention period	The declared lifetime of stored information
Deletion boundary	The age at which removal applies
Residual copy	A remaining duplicate in another store

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Data retention and deletion — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Record ages are 6, 7, 9 days. Cache contains copies of all three.	Toy policy retains session content for 7 days and deletes it when age is at least 7.	Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

By the end, I can	How I will show it
Explain the mechanism and terminology of data retention and deletion	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.
- Which labelled step matches this detail? The scoped deletion passes only after both T2 copies are removed.
- 1. Check age
- 2. Find copies
- 3. Verify removal
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Verify removal The scoped deletion passes only after both T2 copies are removed. Retention requires an age rule and an inventory.

Deleting the visible copy alone does not demonstrate removal of listed backups.

Changed-case question and optional hint

- Changed-case practice: Primary deletion succeeds but backup deletion fails. Which conclusion is supported by the supplied facts?
- 1. Record incomplete deletion and follow the retry or escalation procedure.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

Repair and retain the record

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

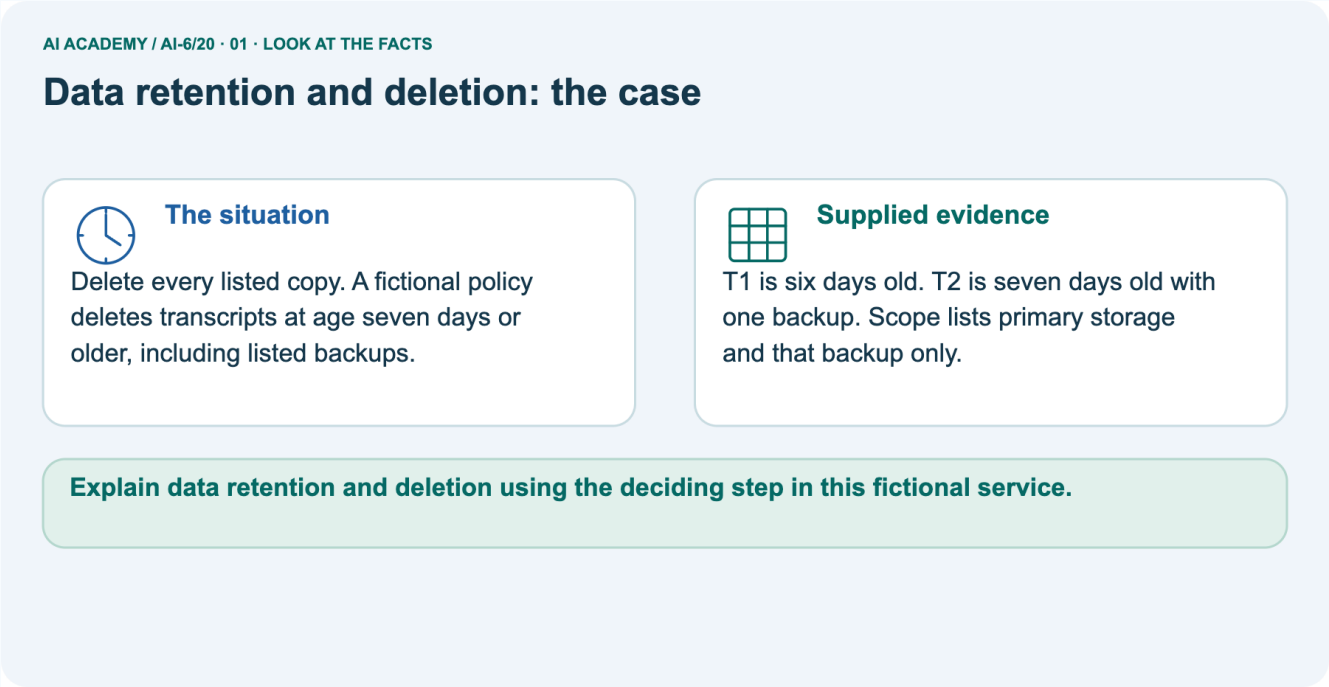
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Data retention and deletion: the case



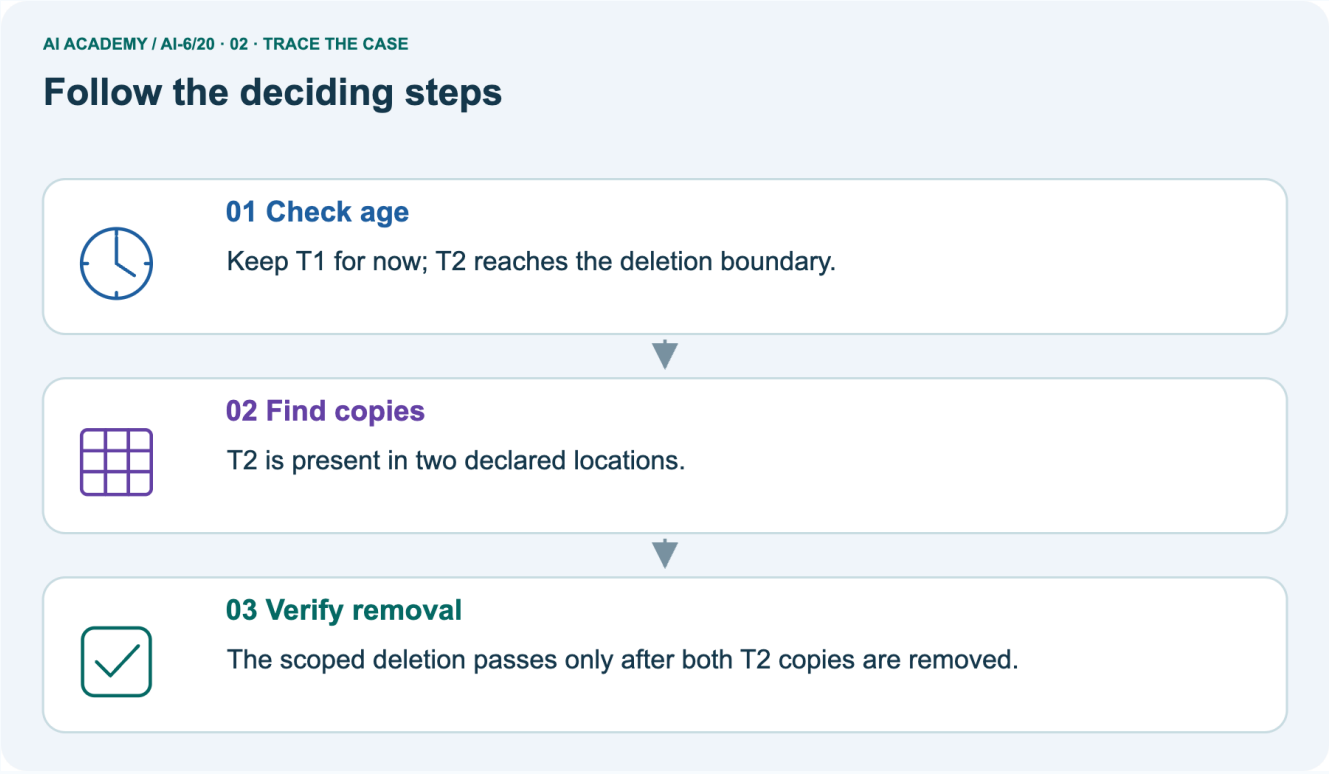
Follow the labelled picture with the matching explanation. Open the vector diagram

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups. Supplied facts: T1 is six days old. T2 is seven days old with one backup.

Scope lists primary storage and that backup only.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

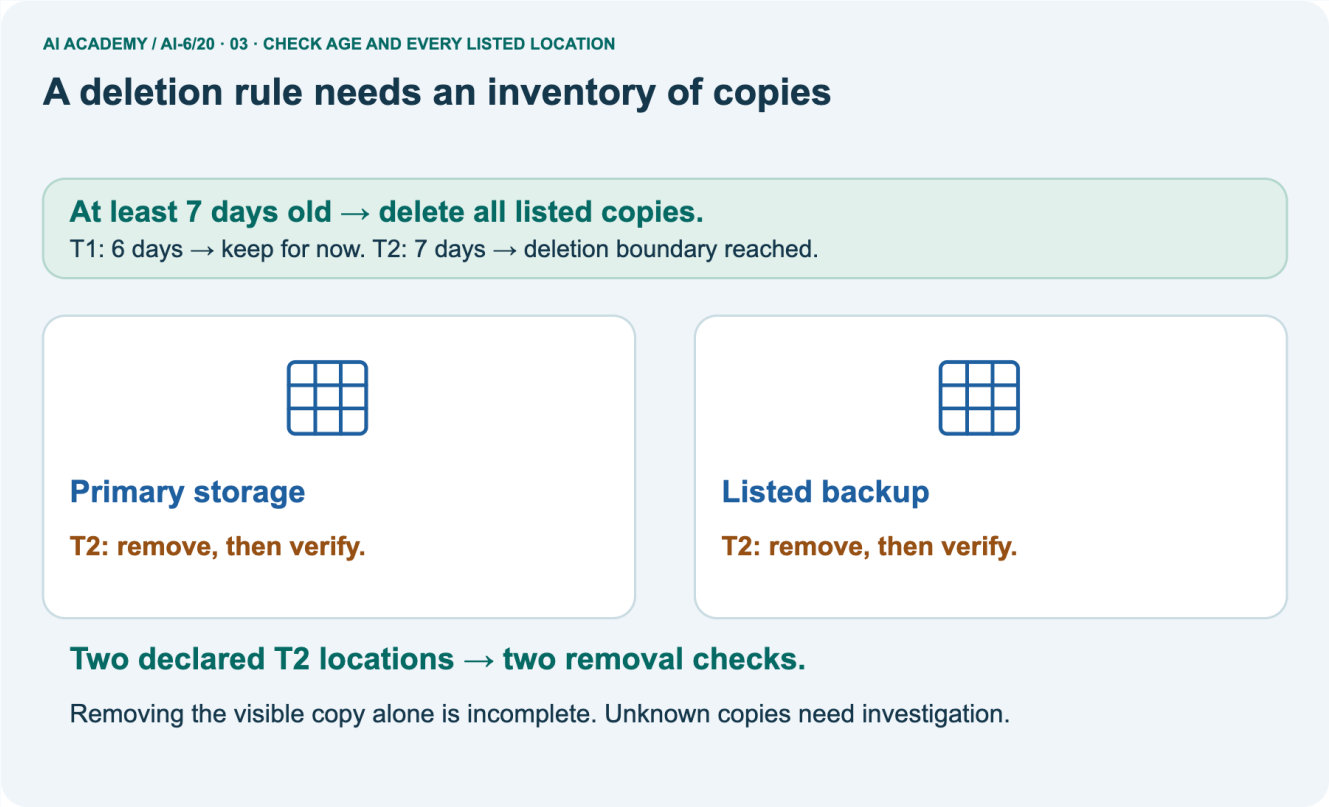


Follow the labelled picture with the matching explanation. Open the vector diagram

Check age: Keep T1 for now; T2 reaches the **deletion boundary**. → Find copies: T2 is present in two declared locations. → Verify removal: The scoped deletion passes only after both T2 copies are removed.

Point to each numbered stage and explain which recorded **input** it uses.

03. A deletion rule needs an inventory of copies



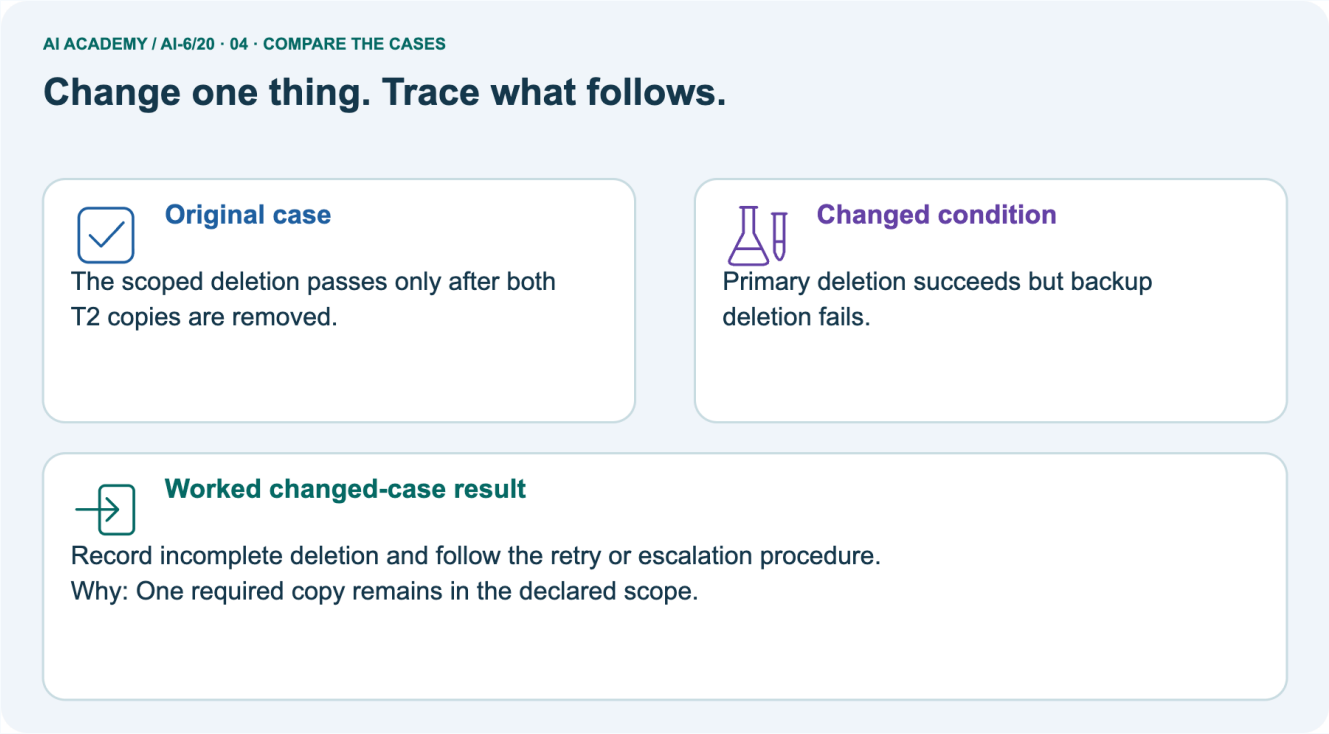
Follow the labelled picture with the matching explanation. Open the vector diagram

The toy retention boundary is seven days or older. Keep six-day T1 for now. Seven-day T2 appears in primary storage and one listed backup; both locations need removal verification.

This diagram specifies required checks, not an actual deletion run.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



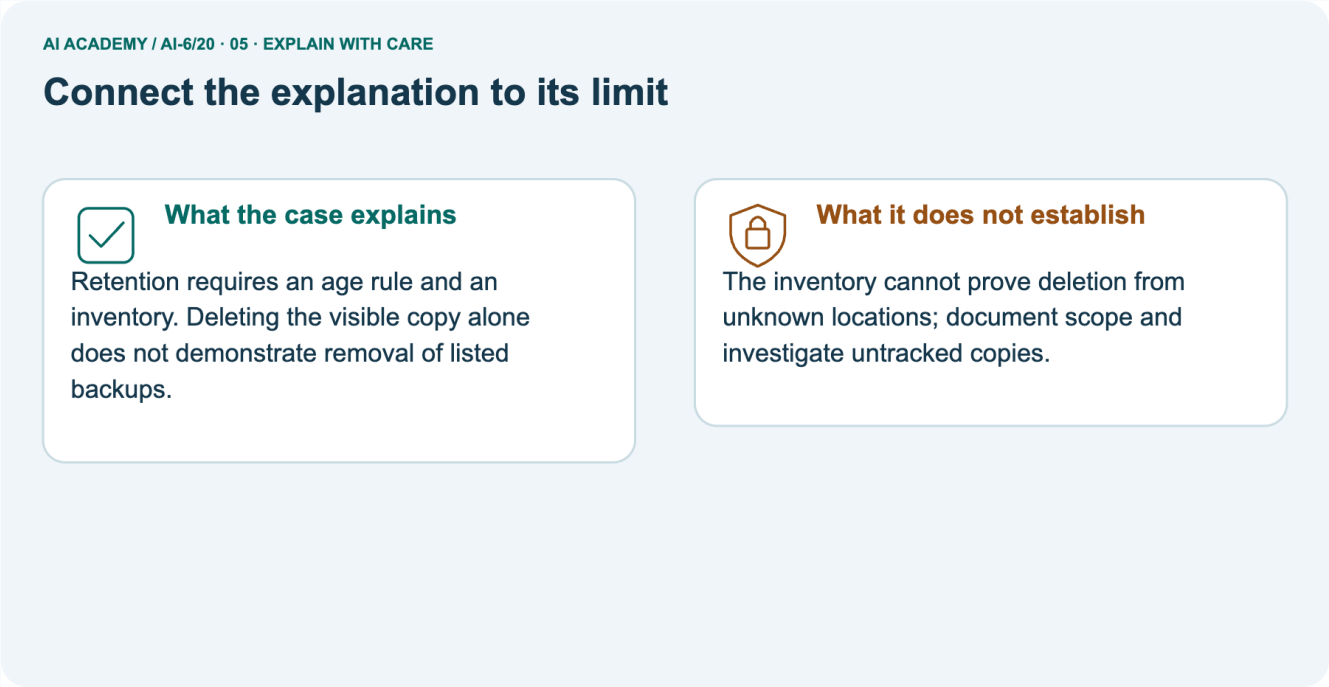
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The scoped deletion passes only after both T2 copies are removed. Changed condition: Primary deletion succeeds but backup deletion fails.

Expected result: Record incomplete deletion and follow the retry or escalation procedure. Reason: One required copy remains in the declared scope.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



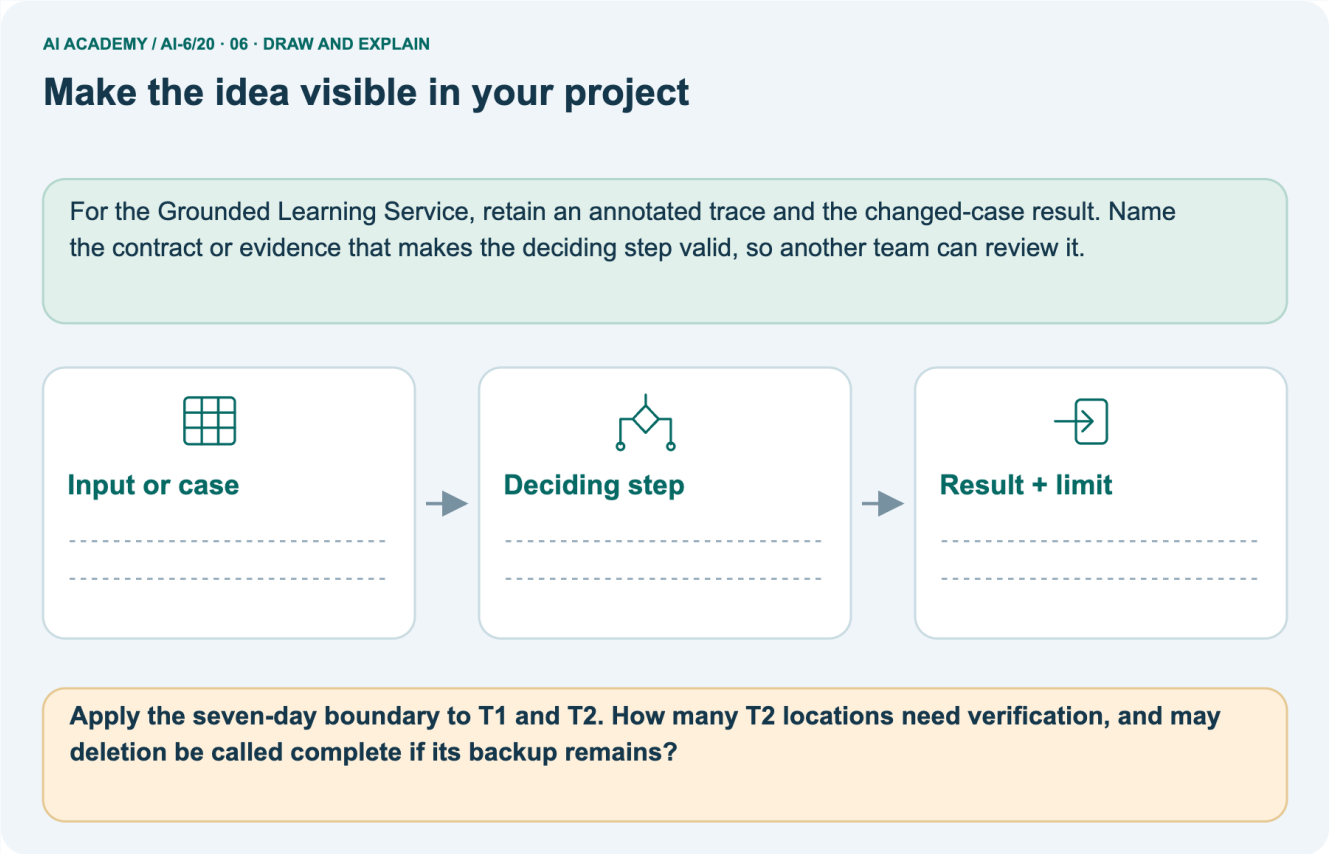
Follow the labelled picture with the matching explanation. Open the vector diagram

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

Limit: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Data retention and deletion: the case

02. Follow the deciding steps

03. A deletion rule needs an inventory of copies

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.